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Group Report

MakeMe

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1. Abstract

Maintaining consistent exercise habits over an extended period can be challenging, particularly when fitness progress is primarily represented through numerical data. MakeMe is a gamified fitness-tracking application designed to address this challenge by integrating personalized avatar progression into the fitness-tracking experience. The application translates users' real-world activities into visual changes in a personalized virtual character, with its initial minimum viable product (MVP) incorporating four core components: character customization, workout logging, muscle-specific progression, and a 90-day game cycle. Character customization allows users' to personalize their avatars, establishing a stronger sense of connection with their virtual representations, while the 90-day cycle introduces a structured progression period that provides a longer-term objective. At the core of this system, workout logging records users' training activity, which drives muscle-specific growth in their avatars. This creates a direct relationship between real-world exercise and virtual character development, allowing users' to visualize how their individual training activities contribute to their avatars progression. Through these interconnected mechanics, MakeMe aims to provide users' with a more engaging representation of their fitness progress and encourage consistent exercise through game-inspired progression mechanics.

2. Introduction

2.1. Background and Context

Conventional fitness-tracking applications provide users with a range of tools to monitor and manage their physical activities. Activities that were traditionally recorded through handwritten logs and manual calculations can now be conveniently organized within digital platforms, allowing users to track workout frequency, monitor performance, and review their progress over time. These features provide valuable feedback and help users establish a structured approach to their fitness routines. Across many of these applications, numerical statistics remain a central method of presenting users' progress. However, while these measurements provide useful information, they offer limited opportunities for users to visually identify with their progress or develop an aspirational connection with their fitness goals. As a result, the ability to monitor fitness activity does not necessarily translate into sustained motivation or consistent exercise habits.

One potential approach to making fitness progress more engaging is through the use of personalized virtual avatars. These avatars can serve as visual representations of users, gradually developing alongside their real-world physical activities. By allowing users to customize their virtual characters, applications can create opportunities for stronger self-identification, making the avatar's development more personally meaningful. Furthermore, allowing these characters to develop toward an idealized physique introduces an aspirational element, giving users a visual representation of what they may hope to achieve through their own fitness journey. Through this combination of self-representation and aspiration, virtual avatars can provide users with a more personal and engaging connection to their fitness goals.

Building upon these concepts, MakeMe is a gamified fitness-tracking application designed to connect users' real-world exercise activities with the development of a personalized virtual avatar. The initial minimum viable product (MVP) focuses on four core components: character customization, workout logging, muscle-specific progression, and a 90-day game cycle. Users can personalize their characters and record their workout activities, which contribute to the gradual development of specific muscle groups within the avatar. Rather than representing physical development as a generalized progression, MakeMe aims to reflect users' individual training habits through muscle-specific visual changes. This progression is further structured

around a 90-day cycle, providing users with a defined period in which to develop their avatars and work toward their fitness goals. By combining workout tracking, character customization, and visual progression, MakeMe seeks to create a more personalized and game-inspired approach to monitoring fitness development.

2.2. Motivation and Problem Statement

Despite the availability of fitness-tracking applications, maintaining consistent engagement with exercise routines remains a challenge for many individuals. Recording workouts and monitoring physical activity can help users organize their fitness routines, but these activities alone do not necessarily provide a compelling reason to remain engaged over an extended period. There is an opportunity to explore how game-inspired progression systems can make continued participation more meaningful by connecting users' exercise efforts with a sense of personal development. The challenge addressed by MakeMe is to design an application that translates individual workout activities into personalized, muscle-specific avatar progression while providing a structured experience through its 90-day game cycle.

The motivation behind developing MakeMe stems from the opportunity to make fitness tracking more engaging through concepts commonly found in video games. By incorporating personalized character development into the fitness-tracking experience, the project explores how users can establish a stronger connection with their fitness goals through virtual self-representation and aspirational progression. Rather than focusing exclusively on recording physical activities, MakeMe aims to provide users with a visual representation of their training efforts, where individual workouts contribute to the gradual development of a character they have personalized. This approach seeks to make fitness progression more personally meaningful while introducing game-inspired mechanics that may encourage users to remain engaged with their exercise routines. Ultimately, the project aims to explore how principles of character customization and progression can be applied to fitness applications to support a more engaging approach to exercise tracking.

2.3. Project Goals and Objectives

MakeMe's primary goal is to explore how conventional fitness-tracking functionality can be extended through game-inspired character progression. By incorporating personalized virtual representations into the fitness-tracking experience, the project aims to establish a more engaging approach to visualizing users' fitness development. The initial MVP serves as the foundation for this broader vision, establishing the essential functionality and progression systems upon which additional features can be developed and integrated in the future.

To establish this foundation, the project pursues the following objectives.

1. Develop a Personalized Avatar System: Implement a customizable virtual character whose appearance can be personalized by users with individual muscle groups that develops independently through recorded workout activities.
2. Integrate Core Fitness-Tracking Functionality: Implement essential workout-logging functionality that allows users to record their exercise activities serving as the foundation for the application's avatar progression system.
3. Establish a Structured Progression System: Develop a 90-day progression cycle that provides users with a defined period to track their activities, observe avatar development, and work toward longer-term fitness goals.

3. Literature Review and Related Work

3.1. Relevant Resrearch and Existing Products

MakeMe's avatar system draws on virtual self-modeling through vicarious reinforcement by Fox and Bailenson, which explored how observing virtual representation of oneself receiving reinforcement would influence exercise behavior [1]. However, their research used virtual representation that resembled the participants, whereas MakeMe features fantasy-based character development that would later skew away from the users physical appearance. This distinction is intentional, as MakeMe allows users to watch their avatars develop into increasingly muscular and powerful characters, potentially inspiring them to aspire to the strength and appearance of their virtual counterparts, motivating them to maintain healthy habits. This design aligns the concept of wishful identification mentioned in the review research by Praetorius and Gorlich on the Proteus Effect, which describes how an avatar's characteristics may influence a users behavior and self-perception [2].Drawing on these concepts, MakeMe's avatar system is designed around two complementary forms of progression: reflective and aspirational. Where the development of the avatar is a direct reflection of users actions, while aspirational progression allows the avatar to evolve towards an idolized physique that users may wish to achieve themselves. While existing research does not directly establish a link between avatar progression and the development of healthy habits, incorporating them may still offer motivational benefits compared with health tracking applications without an avatar.

Although ranking of the top 10 fitness trackers may vary due to subjective reviews and potential biases, many of them can be broadly classified into three categories based on their approaches to fitness tracking and gamification.

Category 1 consists of conventional fitness trackers that incorporate little to no gamification, focusing primarily on workouts, activity logging, and social features [3], [4]. Their popularity suggests that stand-alone fitness monitoring remains an established approach to helping users track their progress and maintain their fitness routine. However, this category represents only one approach to fitness tracking and does not establish whether users would benefit from engaging or game-oriented alternatives.

Category 2 consists of challenge-based gamified fitness trackers that are oriented to encourage physical activity through step-count goals, competitions, social challenges, and real-world incentives. In 2020, (Neupane et al.) reviewed 103 gamified fitness tracker apps and found that challenges, competition and social influence frequently appeared together, while social features played a central role across nearly every identified game genre [5]. These findings demonstrate how conventional activity tracking can be enhanced with game mechanics that encourage users to remain active through external rewards, social interaction, and competition. However, this approach primarily builds on existing fitness tracking activities rather than making exercise part of a broader, evolving game experience.

Category 3 consists of stand-alone games that integrate physical activity directly into gameplay, encouraging users to exercise as they pursue in-game objectives. Games such as PokemonGO and Wii Sports demonstrate the popularity of this approach making physical activity more engaging through exploration and interactive gameplay [6], [7]. In these games, physical avtivity is not simply a means of tracking progress but an integral part of the game itself. Players are motivated to move, explore, and engage with their environment to achieve in-game goals. However, research on PokemonGO suggests that increased in physical activity may be short-lived, highlighting the challenge of sustaining user engagement and long-term exercise habits.

3.2. Comparison With the Proposed Project

Initially, MakeMe was envisioned as an application that would combine elements from the three approaches discussed above, bringing together conventional fitness tracking, gamification, and interactive gameplay. However, as the project developed, its scope was refined to focus on a more specific aspect of the fitness-tracking experience: connecting users' workout activities with the development of their virtual avatars. Rather than attempting to incorporate the various challenges, social features, and gameplay mechanics found in existing solutions, the current MVP places its emphasis on workout logging and muscle-specific avatar progression. Through this approach, recorded exercise activities directly contribute to the development of corresponding muscle groups within a personalized character. This shift allows MakeMe to concentrate on its central design concept providing users with a visual representation of their training efforts through an avatar that develops alongside their fitness journey.

4. Project Requirements

4.1. Product Vision and Scope

4.2. High-Level Requirements

4.3. Target Users and 2 to 5 personas

Intended Users
MakeMe is primarily intended for individuals who enjoy video games and want to establish healthier daily habits through an interactive and rewarding experience. Potential users are limitless though it is mainly targeted to individuals who find conventional fitness applications mundane or difficult to maintain.

5. System Design

5.1. System Architecture

5.2. Use Case Diagram

5.3. UML Class Diagram

5.4. Sequence Diagram

5.5. Design Patterns and Justification for Their Use

6. Implementation and Proof of Concept

6.1. Tools and Technologies

6.2. Implementation of the Main Features

7. Required Resources and Feasibility

7.1. Hardware, Software, and Other Resources

7.2. Feasibility Within the Avaialble Time and Resources

8. Results and Discussion

8.1. Proposed Features and Their Completion Status

8.2. Testing and Results

8.3. Challenges, Limitations, and Reasons for Incomplete Features

9. Conclusion and Future Work

9.1. Summary of Achievements

9.2. Possible Improvements and Extensions

10. References

11. Implementation Goals and Scope

11.1. Implementation Goals

11.2. Core Functional Requirements

- **R1.** Initial Avatar Set Up. The system will collect initial user data through a survey and use the response as the avatars baseline.
- **R2.** Workout Logging and Muscle-Specific Growth. A dedicated workout logger that collects duration, selected exercised body parts, and intensity level from user input. The system uses this input to calculate growth for selected body parts.
- **R3.** Workout Streak Tracking: The system maintains a workout streak using a 48-interval after a workout log, allowing users to include rest days without automatically breaking their streaks.
- **R4.** Avatar Decay. The avatar decays continuously and can be accelerated through poor meal logging habits. This system balances growth from workout logging to simulate a semi-realistic growth rate.
- **R5.** Avatar State System: The system updates the avatars' appearance and state based on the user's logged habits.
- **R6.** Habit Logging. Allows users to record meals, hydration and sleep that contributes to avatars state system and progression.
- **R7.** Bulk Progression. Accumulated throughout the cycle and reads through macro report from macro tracker and calculates progression in clean bulk and dirty bulk that influence the avatars overall appearance.
- **R8.** Recovery and Revival. Prolonged period of inactivity can result to the avatars death. A recovery system detects the users return within three stages that determines recovery state from initial return work out logging to a three-day workout streak. The avatar loses all its progress and will return to its initial state.
- **R9.** Cycle Completion and Summary. The system tracks a 90-day cycle to simulate a three-month transformation and presents a summary of the user's activity, avatar development, and achievements.
- **R10.** Ascension System. Determines if the user has completed their long-term goals and allows the user to begin a new cycle, including a new initial survey and avatar baseline. A badge or title will be rewarded with each ascension.
- **R11.** Limited rewards such as badges or cosmetic items for completing certain milestones.

11.3. Initial Scope

In scope. The initial version of MakeMe will focus on implementing the avatar's initial configuration, muscle-specific growth, workout logging, streak system, and time-based decay. It will also include habit logging, supporting avatars conditional states and bulk progression. A 90-day progression cycle, cycle-completion and summary that provides the user a long term objective.

Out of scope. The initial implementation will exclude extensive equipment collections, multi-player features, competitive leaderboards, and advanced social systems. Character customization, avatar appearance states, and visual progression will also be limited to a predefined set of options. Additionally, the application will not provide medical diagnoses or claim that avatar development accurately represents a user's real body composition or physical health.

12. Initial Design

12.1. Use-Case Diagram

The primary actor is the User, who interacts with the application to track health habits and monitor personal progress. The user's goals include creating an initial avatar, logging health related-activities, viewing the avatars visual development as it changes in stature, appearance, and condition in response to their habits, reviewing macronutrient summaries and cycle progress, avatar personalization, and managing health goals. These use cases illustrate how MakeMe combines health habit tracking with visual character progression to encourage users to maintain healthy routeines throughout the 90-day progression cycle represented in figure 1.

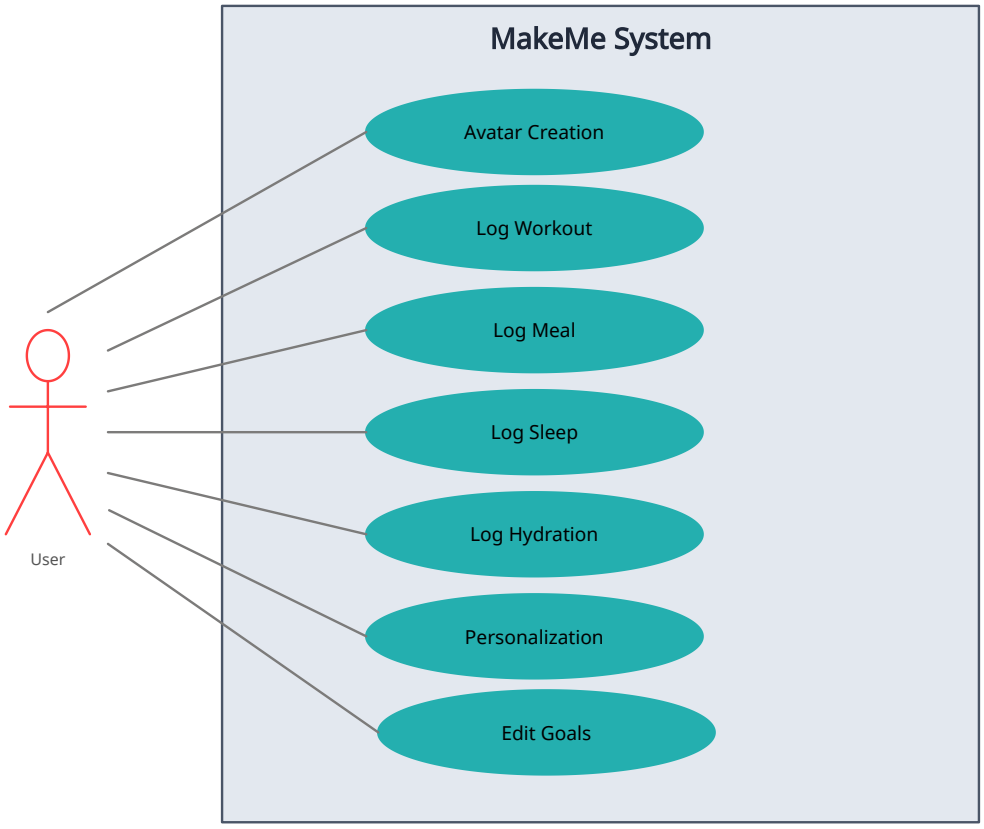


Figure 1: Initial use-case diagram for the proposed system.

12.2. Class Diagram

The class diagram shown in figure 2 organizes MakeMe into classes responsible for user profiles, avatar progression, workout logging, nutrition tracking, and health habits. The Avatar and BodyPart classes manage visual development, while BulkProgression and the streak classes track consistency and progress. The CycleManager manages the 90-day progression cycle, and the RewardSystem handles milestone rewards and equipment unlocks. Relationships between classes primarily use associations and composition, while inheritance connects the abstract Streak class to WorkoutStreak and MealStreak. Enumerations define predefined states and options used throughout the system.

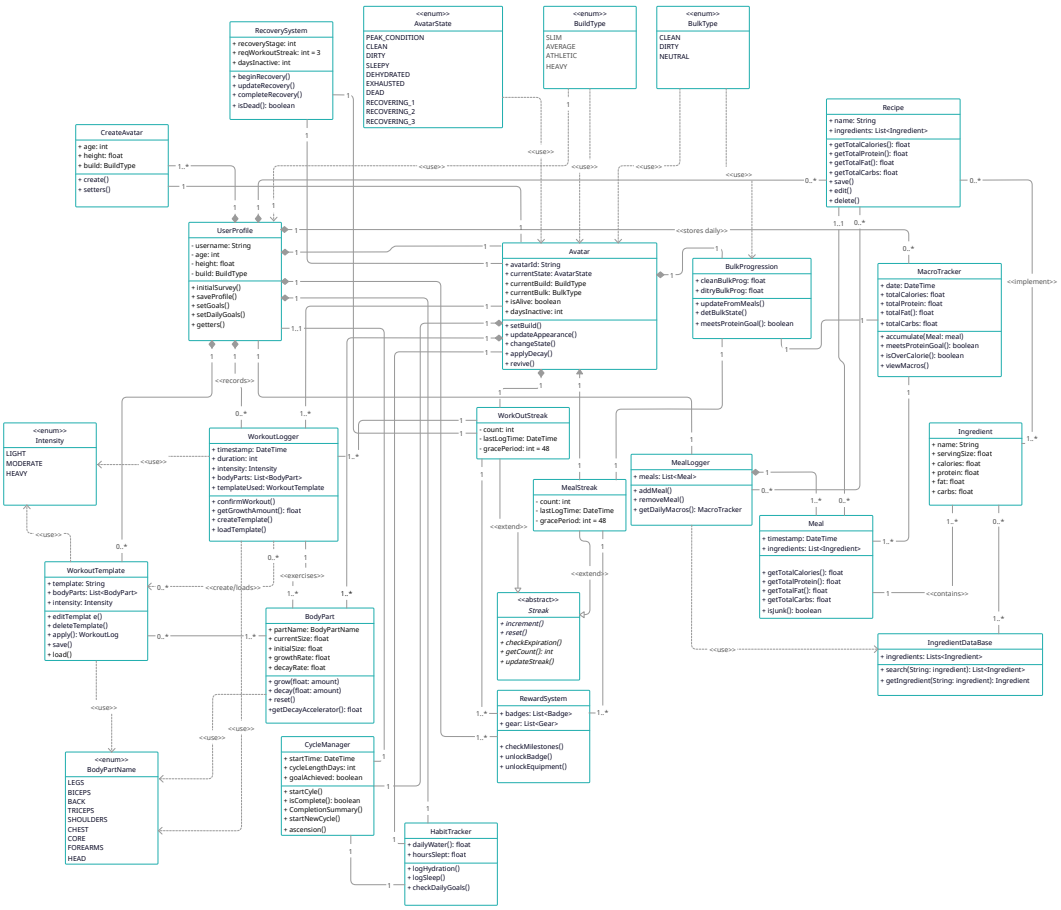


Figure 2: Initial UML class diagram for the proposed system.

13. Feasibility and Supporting Evidence

The technical feasibility of MakeMe depends primarily on developing its avatar progression system, which is expected to require the largest portion of development time. Unlike conventional fitness applications that display numerical statistics, MakeMe aims to visually represent progress through body-part-specific growth. The system must translate workout duration, intensity, targeted muscle groups, consistency, and inactivity into visible changes to the avatar's chest, arms, shoulders, back, core, and legs. To reduce complexity, the initial implementation will use a predefined 2D or semi-2D character model with limited growth stages for each body part, avoiding complex real-time 3D deformation. The project will also exclude multiplayer features, competitive leaderboards, extensive equipment collections, advanced social systems, and unrestricted character customization. Additionally, the application will not provide medical diagnoses or claim that avatar development accurately represents a user's real body composition or physical health. These exclusions allow development to focus on the core features of habit tracking, avatar progression, and the 90-day cycle. An AI generated UI mockup by ChatGPT can be used to visualize the interface and avatar progression, providing a reference for development and user testing as shown in Figure figure 3.

For nutrition tracking, MakeMe could integrate an external API such as USDA FoodData Central[8] to retrieve food information and calculate calories and macronutrients, reducing the need to create a comprehensive food database from scratch. However, the team must account for API integration, serving-size matching, and secure API key management; a smaller predefined database could serve as a fallback. Flutter is a potential framework for developing both Android

and iOS applications from a shared codebase, although learning Dart and Flutter introduces an initial learning curve. Previous experience with Java should help because both languages support object-oriented programming, classes, methods, and familiar control structures. Development can initially focus on Android using Android Studio's emulator to test the interface and core functionality without requiring a physical device. However, iOS development and testing will still require access to the appropriate Apple development tools. Overall, MakeMe is feasible if the team prioritizes a functional minimum viable product, tests the avatar growth and decay rules carefully, and postpones nonessential features until the core tracking and progression systems work reliably.



Figure 3: Supporting feasibility evidence for the proposed project.

14. Resources and Risks

Table 1: Initial project risks and mitigation strategies.

Risk / Constraint	Why it matters	Mitigation / Fallback
Replace with risk 1	Explain the impact	Explain how the risk could be reduced or handled.
Replace with risk 2	Explain the impact	Explain how the risk could be reduced or handled.

References

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[3] A. Hall. "The 10 best workout and fitness apps of 2026," Forbes Health, Accessed: Oct. 12, 2026. [Online]. Available: https://www.forbes.com/health/weight-loss/best-fitness-apps/

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